AWS Certified Solutions Architect

Associate C03

Module 1

Introduction to Amazon Web Services

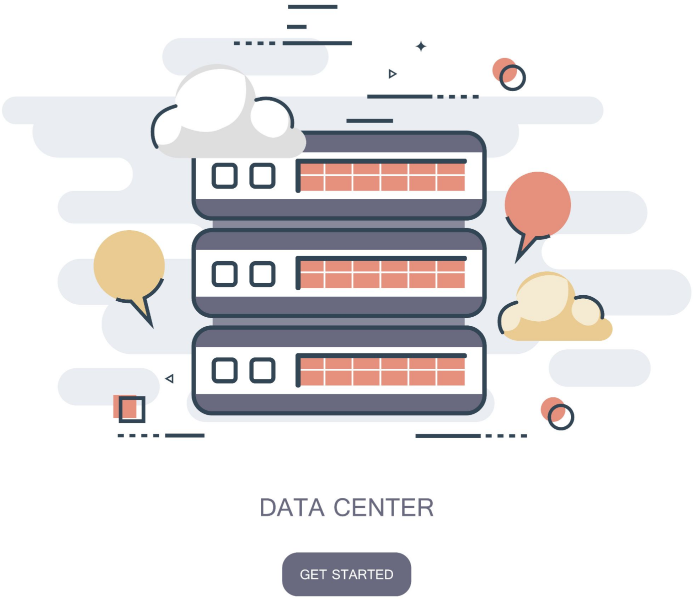
Agenda

1. Introduction to Cloud Computing
2. Cloud Deployment and Service Models
3. Virtualization and Hypervisors
4. AWS Suite (with Analogy)
5. AWS Well-Architected Frameworks

Introduction to Cloud Computing

What Is Cloud?

The **cloud** is a distributed collection of **servers** that host software and infrastructure, and it is **accessed over the Internet**. The cloud can provide services over networks, i.e., on **public networks** or on **private networks**, i.e., WAN, LAN, or VPN. Applications such as e-mail, web conferencing, and customer relationship management (CRM) all run in the cloud.



What Is Cloud Computing?

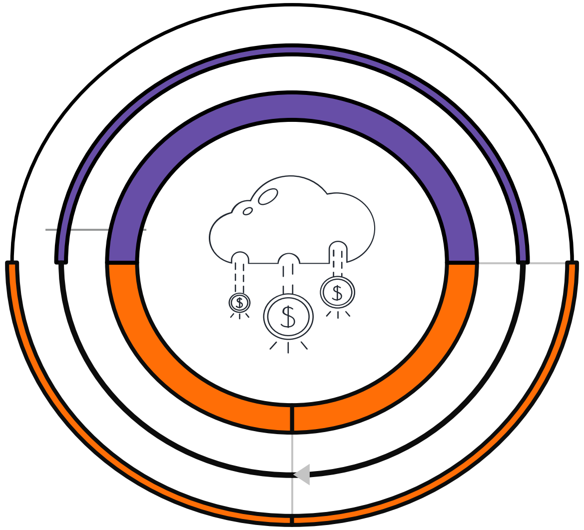
Cloud computing is like renting compute power and storage online. Instead of owning and maintaining your own hardware, you access servers and services over the internet. It's cost-effective, scalable, and allows you to use resources as needed, making it popular for businesses and individuals alike.

Cloud Deployment and Service Models

What Is a Cloud Deployment Model?

A **cloud deployment model** essentially defines where the infrastructure for your **deployment resides** and determines who has **ownership** and control **over that infrastructure**. It also determines the cloud's nature and purpose.

Types of Cloud Deployment Models



Public Cloud

(Shared resources open to everyone)

Private Cloud

(Dedicated resources of organizations)

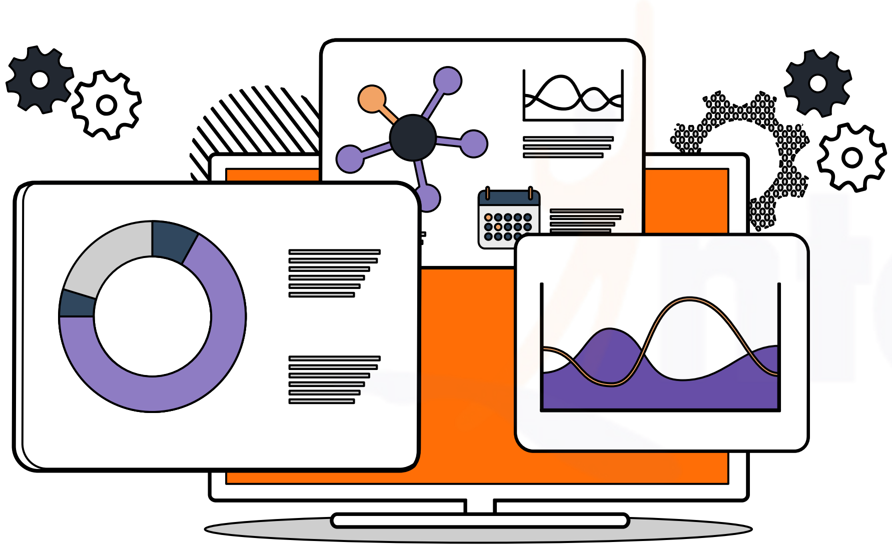
Hybrid Cloud

(A mix of public and private)

What Is a Cloud Service Model?

There are **three main** types of service models in cloud computing. Each type of cloud computing provides different levels of **control**, **flexibility**, and **management**, so you'll select the proper set of services for your needs.

Types of Cloud Service Models



IaaS

Infrastructure as a Service

PaaS

Platform as a Service

SaaS

Software as a Service

Understanding the Service Models

On-site	IaaS	PaaS	SaaS
Applications	Applications	Applications	Applications
Data	Data	Data	Data
Runtime	Runtime	Runtime	Runtime
Middleware	Middleware	Middleware	Middleware
O/S	O/S	O/S	O/S
Virtualization	Virtualization	Virtualization	Virtualization
Servers	Servers	Servers	Servers
Storage	Storage	Storage	Storage
Networking	Networking	Networking	Networking

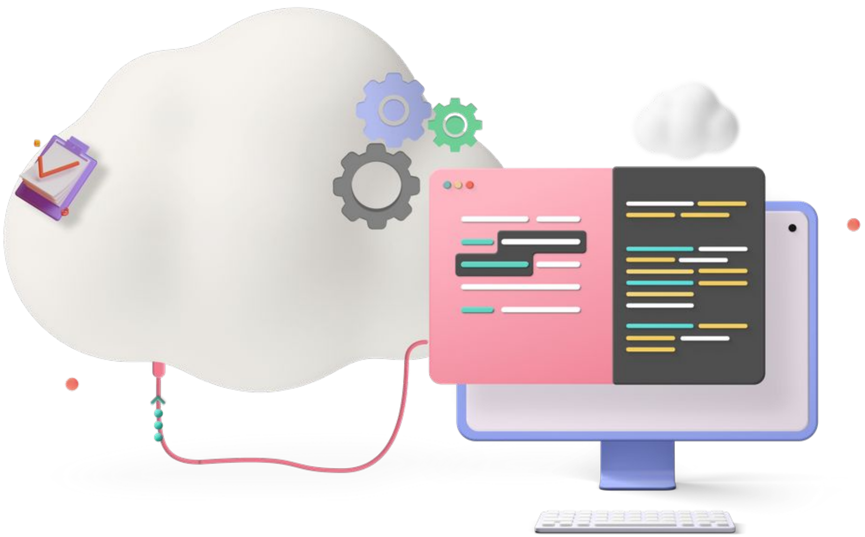
You manage

Service provider manages

Virtualization and Hypervisor

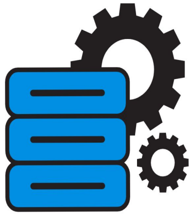
What Is Virtualization?

Virtualization is a technology that creates a **simulated** or **virtual version** of computer **hardware**, **software**, **storage**, or **networks**. It allows multiple operating systems or applications to run on a single physical machine, improving **efficiency**, **resource utilization**, and **flexibility** in IT infrastructure.



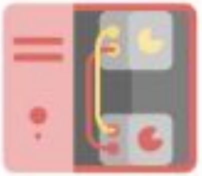
NOTE: Virtualization can exist without Cloud, but Cloud Computing **cannot** exist without Virtualization.

Types of Virtualization



Hardware

Virtualizes computer hardware components.



Storage

Aggregates storage into a virtual pool.

Network

Combine multiple networks into one.

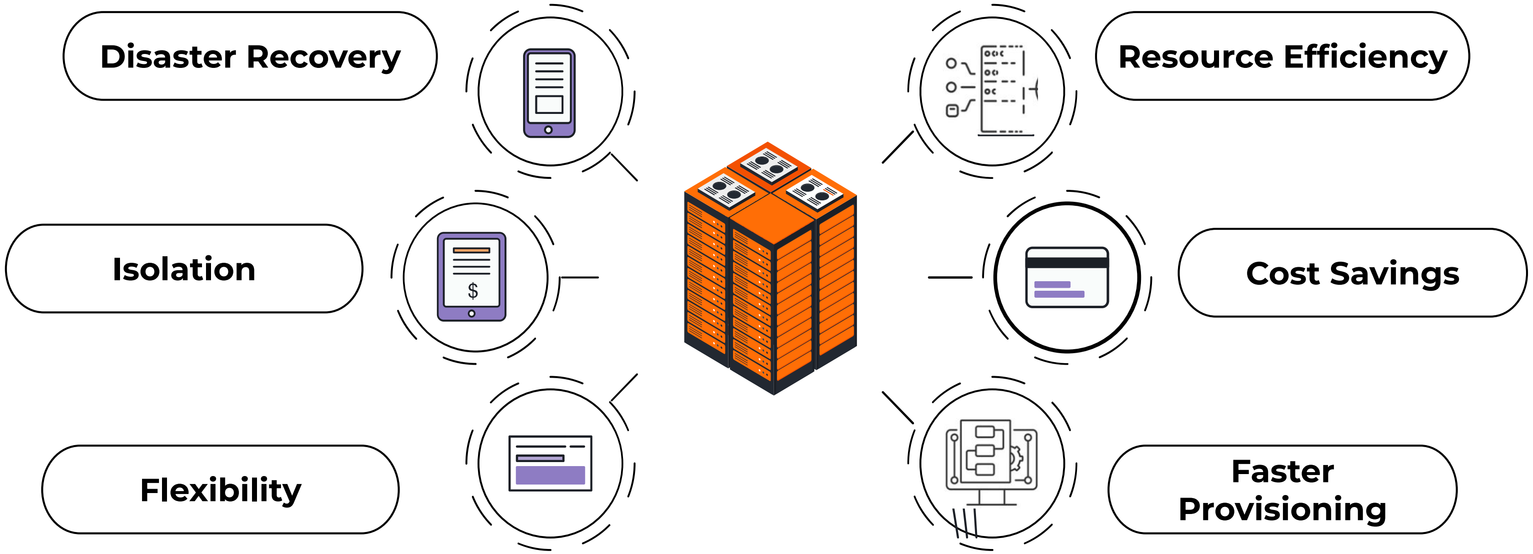
Server

Multiple virtual servers on one

Application

Runs apps independently of OS

Advantages of Virtualization

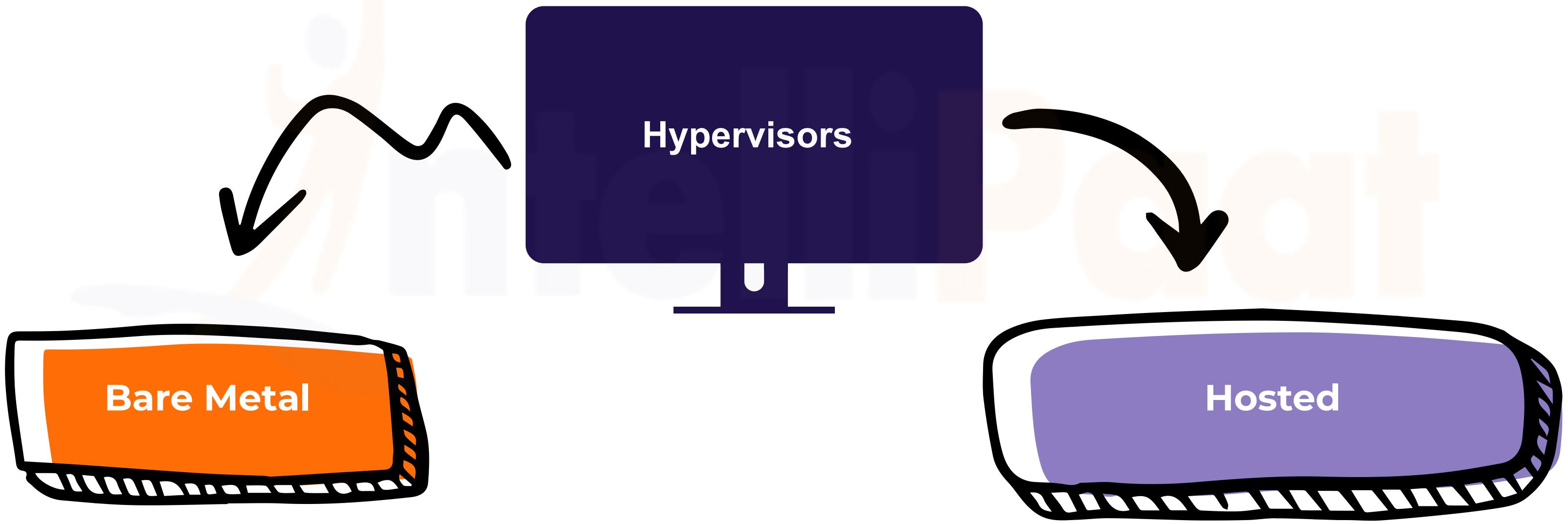


What Is a Hypervisor?

A **hypervisor** is software that enables multiple operating systems to run on a single physical machine simultaneously. It **creates** and **manages** virtual machines, allowing **efficient resource utilization** and **isolation** of different computing environments on the same hardware.

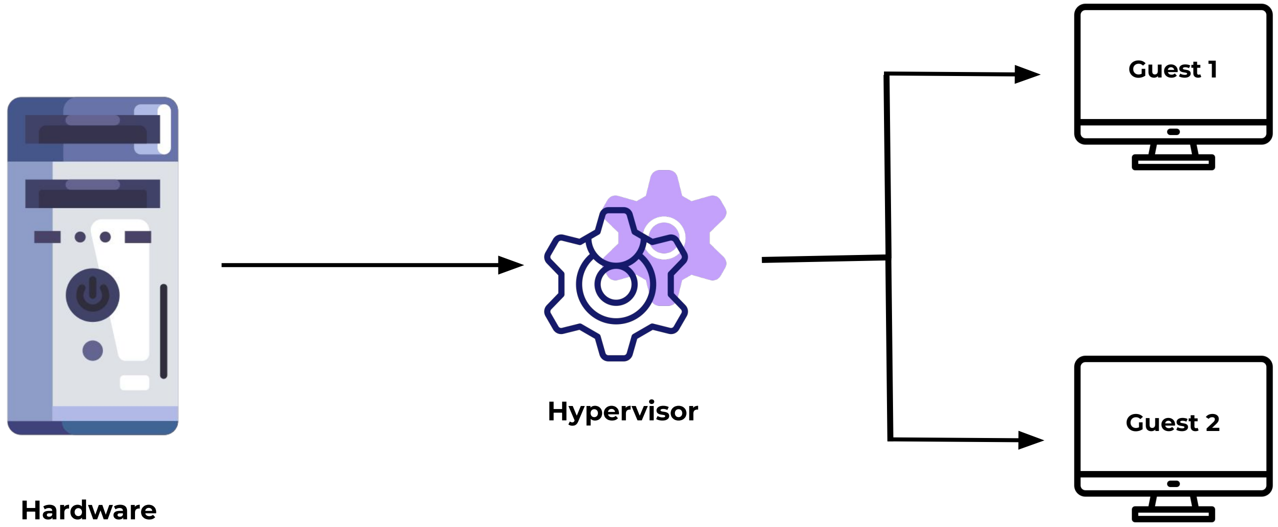
Types of Hypervisors

There are two main hypervisor types, referred to as “**Type 1**” (or “**bare metal**”) and “**Type 2**” (or “**hosted**”).



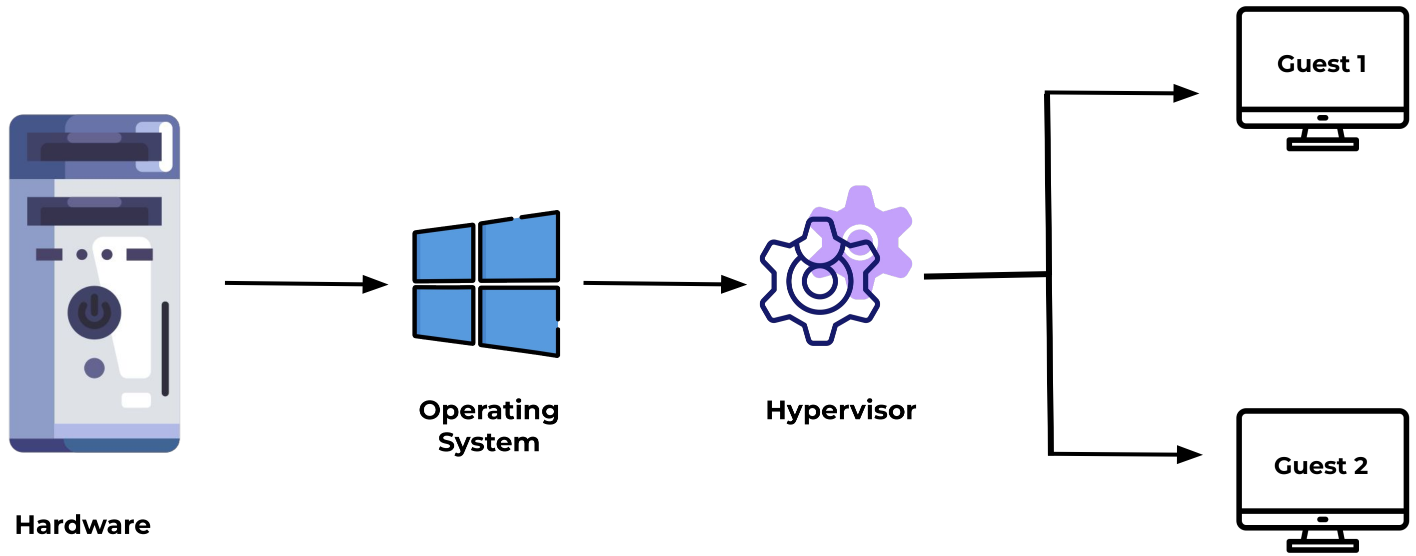
Type 1 - Bare Metal Hypervisor

Type 1 hypervisors, or **bare-metal** hypervisors, install directly on hardware. It optimizes performance by avoiding an underlying OS, making it ideal for production servers where efficiency and resource utilization are paramount.



Type 2 - Hosted Hypervisor

A **Type 2** Hypervisors, or **Hosted** Hypervisors, install on top of an **existing operating system**. It's suitable for development and testing, offering flexibility at the cost of slightly lower performance. Examples include **VMware Workstation** and **Oracle VirtualBox**.

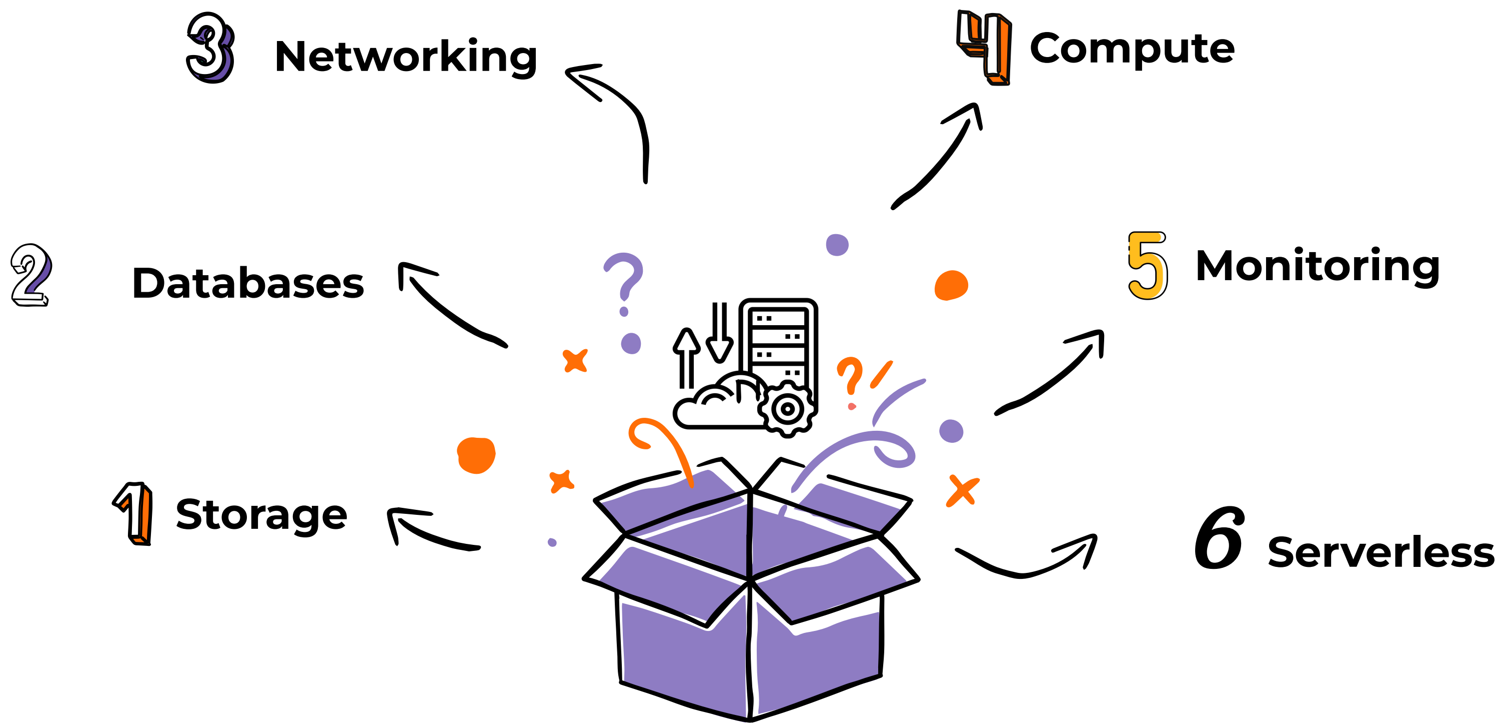


Virtualization in AWS

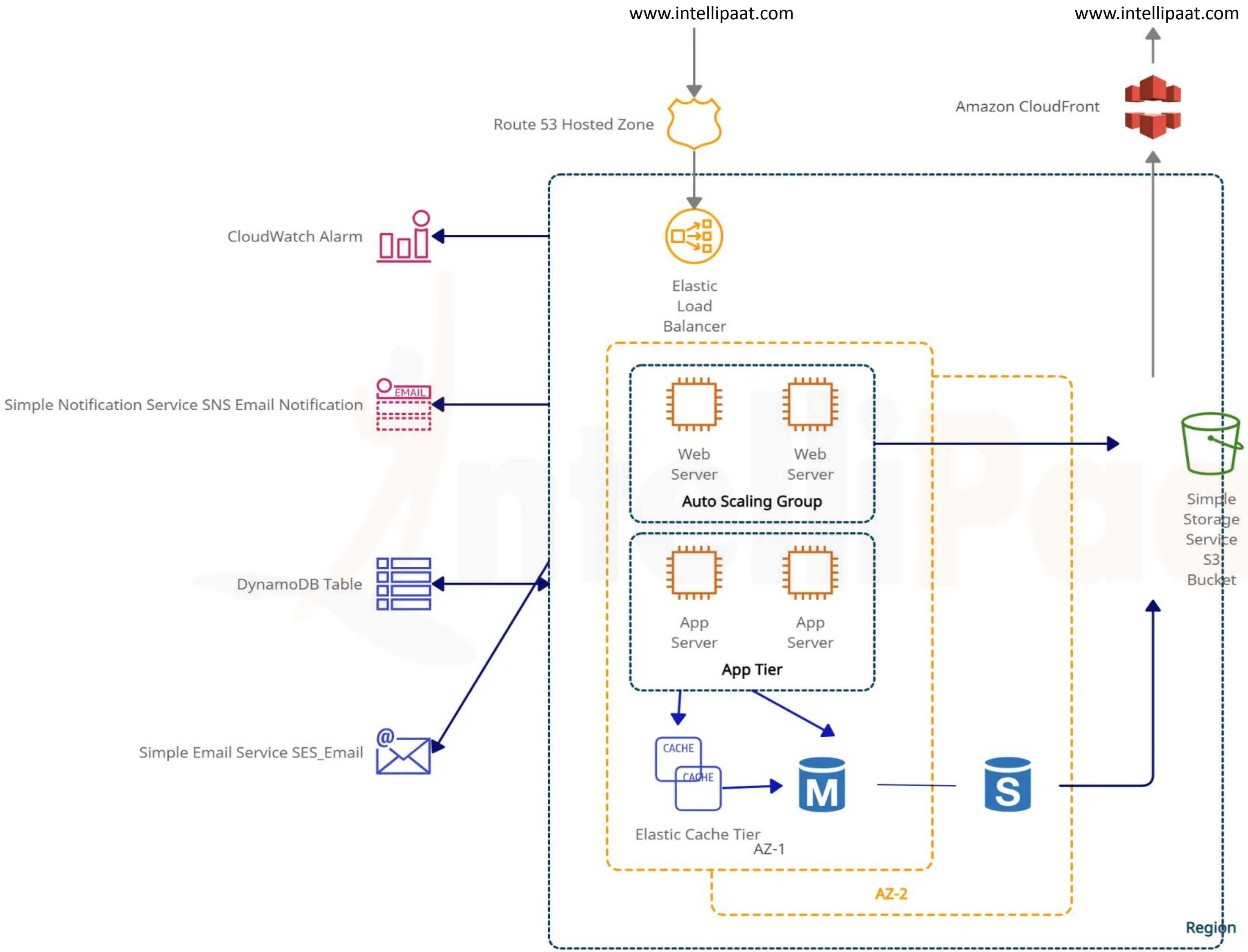
AWS achieves virtualization using the **XEN hypervisor**. Every AWS AMI uses the bare metal Xen hypervisor. Xen offers **two types** of virtualization: HVM (**Hardware Virtual Machine**) and PV (**Paravirtualization**).

AWS Suite

What Does the AWS Suite Consist of?



Case Study: AWS Services Architecture



AWS Well Architected Framework

AWS Well Architected Framework

Operational Excellence

Operational Excellence

- **Optimize** operations, understand processes, and improve continually for value delivery.
- **Guidance** on best practices for DevOps, automation, observability, and infrastructure.
- **Refine** operations procedures frequently.

Security

Security

- The AWS Security pillar **safeguards data**, **systems**, and **assets** through cloud technology.
- The security pillar offers best practices for **identity management** and **infrastructure protection.**
- The security pillar comprises seven design principles **emphasizing identity** and **traceability.**

Performance Efficiency

Performance Efficiency

- Efficiently utilizes resources, meeting demands, and adapting to evolving technologies.
- Emphasizes efficiency, architecture selection, and advanced tech like serverless, machine learning.

Reliability

Reliability

- The reliability pillar assures **consistent workload** performance, **operation**, and **testing**, offering detailed guidance for reliable implementations.

- The pillar stresses robust foundations, resilient design, **change management**, and reliable failure recovery processes.

Sustainability

Sustainability

- The sustainability pillar **minimizes environmental impacts**, focusing on energy efficiency.
- Guidance on shared responsibility for sustainability, resource optimization, and minimizing downstream effects.
- Prioritizes informed decisions **balancing security**, **cost**, **performance**, **reliability**, and **sustainability**.

Cost Optimization

Cost Optimization

- Ensure resources are utilized optimally to **avoid unnecessary expenses.**

- **Prioritize cost** reduction without compromising on delivering business value.

Hands-On

Sign Up on AWS Management Console

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